Digital democracy encompasses many things, from simple digital identity cards in India to e-residency in Estonia. In the context of this chapter, you'll be concerned with systems that enable democracy by using digitization, such as internet-based communication and information systems--in particular, using internet-based electronic voting systems instead of paper ballots or mechanical machines. Let's begin discussing the problem with a problem statement.

> **PROBLEM STATEMENT** Consider an online ballot application. People vote to choose a proposal from a set of proposals. A chairperson registers the people who can vote, and only registered voters can vote (only once) on a proposal of their choice. The chairperson's vote is weighted twice (x2) as heavily as regular people's votes. The ballot process goes through four states (`Init`, `Regs`, `Vote`, `Done`), and the respective operations (initialize, register, vote, count votes) can be performed only in the corresponding states.

3.2.1 Designing a solution

We'll apply the design principles you learned in chapter 2, which are available in appendix B. Please review these principles before starting the design process.

Here are the recommended steps for solving the ballot problem:

- Apply design principles 1, 2, and 3 to design the *use case diagram*; use that diagram to discover the users, data assets, and transactions.
- Use design principle 4, design the *contract diagram* that defines the data, modifiers, or rules for verification and validation, and functions.
- Using the contract diagram, develop the *smart contract* in Solidity.
- Compile and deploy the smart contract in the Remix IDE, and test it.